

Gravitation Notes:

I] Textbook Questions:

1 Mark Objectives:

1. Study the entries in the following table and rewrite them putting the connected items in a single row. [2026]

I	II	III
Mass	m/s^2	Zero at the centre
Weight	kg	Measure of inertia
Acceleration due to gravity	Nm^2/kg^2	Same in the entire universe
Gravitational constant	N	Depends on height

Ans:

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Mass	kg	Measure of inertia
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2. What are

(i) Free fall: **OR** What is free fall? When is it possible? [2024]

When a body falls towards the Earth under the influence of the Earth's gravity alone, its motion is called a 'free fall'. True Free fall is possible only in vacuum.

(ii) Acceleration due to gravity:

The gravitational force due to earth on a body result in its acceleration. This is called acceleration due to gravity. It is denoted by g and given by the formula: $g = \frac{GM}{R^2}$

(iii) Escape velocity:

[2024]

The minimum initial velocity needed by an object projected upwards to overcome Earths gravitational force and not fall back on Earth is called escape velocity. It is denoted by V_{esc} and given by the formula:

$$v_{esc} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}.$$

(iv) Centripetal force:

The force that acts on any object moving along a circle, directed towards the centre of the circle is called centripetal force. It is given by the formula: $F = \frac{mv^2}{r}$.

2 Marks Questions:

Q. Give Reasons:

1. Explain why the value of g is zero at the centre of the earth.

Ans:

1. The acceleration due to gravity is given by the formula $g = \frac{GM}{R^2}$.
2. where R is the radius, G is the gravitational constant and M is the mass of the Earth.
3. At the centre of the Earth, both R and M can be considered to be zero.
4. Therefore, according to the formula the value of g becomes zero at the centre of the Earth.

OR

Ans:

1. The value of g changes if we go inside the earth. The value of R in equation of acceleration due to gravity: $g = \frac{GM}{R^2}$ decreases and one would think that the value of g should increase as per the formula.
2. However, the part of the earth which contributes towards the gravitational force felt by the object also decreases.
3. Which means that the value of M to be used in equation $g = \frac{GM}{R^2}$ also decreases.
4. As a combined result of change in R and M , the value of g decreases as we go deep inside the earth. Ultimately the value of g becomes zero at the centre of the earth.

Q. Answer the following:

1. If the value of ' g ' suddenly becomes twice its value, it will become two times more difficult to pull a heavy object along the floor.

Ans:

1. The weight of an object is defined as the force with which Earth attracts the object.
2. The force (F) on an object of mass m on the surface of Earth is equal to mg .
3. If the value of ' g ' were doubled, the force ($F = mg$) on the object would become twice.
4. As a result, the weight will get doubled. So, it will become two times more difficult to pull a heavy object on the floor.

2. Will the mass and weight of an object on the earth be same as their values on Mars? Why?

Ans:

1. Mass is the amount of matter present in an object which is constant everywhere.
2. Weight is the force with which the earth attracts an object.
3. Weight (W) of an object at a place depends on the acceleration due to gravity of the place i.e. $W = mg$.
4. Since the values of acceleration due to gravity (g) on Earth and Mars is different, thus the weight of the object will be different for both the planets.

Q. Difference between:

1. Mass and Weight

[2023, 2025]

*Any two distinguishing points for 2 marks.

Ans.

Mass	Weight
1. Mass is the amount of matter present in an object.	1. Weight is the force with which Earth attracts an object.
2. SI unit of mass is kg.	2. SI unit of weight is Newton.
3. Mass is a scalar quantity.	3. Weight is a vector quantity.
4. Its value is a constant.	4. Its value varies.

5 Marks Questions:**1. Write the three laws given by Kepler.**

[2020, 2022, 2023]

Ans.

- Kepler's first law:** The orbit of a planet is an ellipse with the Sun at one of the foci.
- Kepler's second law:** The line joining the planet and the Sun sweeps equal areas in equal intervals of time.
- Kepler's third law:** The square of the period of revolution of a planet around the Sun is directly proportional to the cube of the mean distance of the planet from the Sun.

$$T^2 \propto r^3 \quad \text{i.e.} \quad \frac{T^2}{r^3} = \text{constant} = K$$

II] Extra - Previous Years' Questions: (PYQs)**1 Mark Objectives:**

1. The value of escape velocity of a body from the earth's surface is $V_{esc} = \underline{\hspace{2cm}}$ [2026]

(A) $\sqrt{\frac{GM}{R}}$ (B) $2\sqrt{\frac{GM}{R}}$ ans: (C) $\sqrt{\frac{2GM}{R}}$ (D) $\sqrt{\frac{GM}{2R}}$

2. By observing the correlation in the first pair, complete the second pair: [2026]

Mass : Kilogram :: Weight : **Newton**

2 Marks Questions:**Q. Answer the following:**

1. Will the value of 'g' be the same everywhere on the surface of Earth? Justify. [2022]

Ans:

- The shape of the earth is not exactly spherical and so the distance of a point on the surface of the earth from its centre differs somewhat from place to place.
- Due to its rotation, the earth bulges at the equator and is flatter at the poles.

- Its radius is largest at the equator and smallest at the poles.
- The value of g is thus highest (9.832 m/s^2) at the poles and decreases slowly with decreasing latitude. It is lowest (9.78 m/s^2) at the equator.

3 Marks Questions:

Q. Answer the following:

1. Explain the concept 'Escape Velocity' in detail.

[2025]

Ans:

- The minimum initial velocity needed by an object projected upwards to overcome Earth's gravitational force and not fall back on Earth is called escape velocity.
- It is denoted by V_{esc} and given by the formula: $V_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{2gR}$.
- The value of g keeps decreasing as we go higher above the surface of the earth.
- Thus, the force pulling the object downward, decreases as the object goes up.
- If we keep increasing the initial velocity of the object, it will reach larger heights and above a particular value of initial velocity of the object, the object is able to overcome the downward pull by the earth.
- It can escape the earth forever and will not fall back on the earth. This velocity is called escape velocity.

2. Observe the figure and answer the questions:

[2024]

a. State Newton's universal law of gravitation.

Ans: According to this theory every object in the universe attracts every other object with a definite force. This force is directly proportional to the product of the masses of the two objects and is inversely proportional to the square of the distance between them.



b. If the distance between the two bodies is tripled, how will the gravitational force between them change?

Ans: Gravitational force is given by: $F = G \frac{m_1 m_2}{(d)^2}$.

If the distance between the two bodies is tripled, the gravitational force between them would decrease by a factor of 9, because if (d) becomes $(3d)$, then $F = G \frac{m_1 m_2}{(3d)^2} = \frac{G m_1 m_2}{9 d^2}$

c. What will happen to gravitational force, if mass of one of the objects is doubled?

Ans: If the mass of one of the objects is doubled, the gravitational force between them will also double. since m_1 becomes $2m_1$ then, $F = G \frac{2m_1 m_2}{d^2} = 2 \left(G \frac{m_1 m_2}{d^2} \right)$

3. Explain on which factors the value of acceleration due to gravity 'g' depends.

[2024]

Ans:

Variation in the value of g:

1. The acceleration due to gravity (g) depends on the mass (M) and the radius (R) of the earth, given by the formula: $g = \frac{GM}{R^2}$
2. **Change along the surface of the earth:** the value of (g) is not constant across the earth's surface because earth is not a perfect sphere.
3. Due to earth's rotation, it bulges at the equator and is flattened at the poles, causing variation in radius of earth (R).
4. Due to this the value of g is maximum at poles $\approx 9.832 \text{ m/s}^2$ and minimum at equator $\approx 9.78 \text{ m/s}^2$
5. **Change with height:** as height above earth increases, distance from the centre increases, so (g) decreases. However, for small heights (like airplanes), the change is negligible, but for satellites, it is significant.
6. **Change with depth:** as we go inside earth the effective mass and radius contributing to gravity decreases. As a combined effect, (g) decreases continuously and it becomes zero at the centre of the earth.

